

# The Potential-Energy Hinge

Configuration, interaction and the chosen zero

IB Physics A3 worksheet

**Central idea.** Potential energy belongs to a configuration of interacting objects. Its expression depends on the interaction law and on the chosen reference level.

$$\Delta E_p = -W_{\text{conservative force}}.$$

In each panel, identify: **(i)** the interacting objects, **(ii)** the configuration variable, **(iii)** the chosen zero of energy, and **(iv)** the sign of the energy change.

## 1. Near-Earth gravitational energy

For a uniform gravitational field,

$$\Delta E_{p,g} = mg\Delta h.$$

A ball of mass 0.40 kg moves from height  $h_1 = 1.2$  m to  $h_2 = 4.7$  m.

**Tasks**

1. Calculate  $\Delta E_{p,g}$ .

2. State the work done by gravity during the motion.

3. If the zero of potential energy is moved upward by 2.0 m, what changes and what remains unchanged?

4. Explain why  $mgh$  is a local expression rather than a universal gravitational formula.

**Hinge question** Which part is physics and which part is bookkeeping: the value of  $g$ , the height difference, or the chosen zero?

## 2. Elastic potential energy

For an ideal spring,

$$E_{p,s} = \frac{1}{2}kx^2,$$

where  $x = 0$  is the unstretched position.

A spring of stiffness  $k = 180 \text{ N m}^{-1}$  is stretched from  $x_1 = 0.04$  m to  $x_2 = 0.11$  m.

**Tasks**

1. Calculate the change in elastic potential energy.

2. State the work done by the spring during this change.

3. Explain why the energy depends on  $x^2$  although the restoring force depends on  $x$ .

4. State the condition under which this expression is valid.

**Hinge question** What supplies the closure here: energy conservation, Hooke’s law, or the choice  $x = 0$ ? Explain the role of each.

## 3. Radial fields and signs

For two point masses,

$$E_{p,g} = -\frac{GMm}{r},$$

with  $E_{p,g} = 0$  at  $r \rightarrow \infty$ .

For two point charges,

$$E_{p,e} = \frac{kQq}{r},$$

with  $E_{p,e} = 0$  at  $r \rightarrow \infty$ .

**Tasks**

1. Explain why gravitational potential energy is negative with this reference.

2. State when electric potential energy is positive and when it is negative.

3. A satellite moves from  $r$  to  $2r$ . Is its gravitational potential energy larger or smaller? Explain without calculation.

4. Two positive charges move farther apart. State the signs of  $\Delta E_{p,e}$  and of the work done by the electric force.

**Hinge question** Why can two expressions with the same  $1/r$  shape carry different signs?

**Synthesis.** For each case, write the configuration variable and the physical law that supplies the expression:

|                      |  |  |
|----------------------|--|--|
| near-Earth gravity   |  |  |
| spring               |  |  |
| radial gravity       |  |  |
| electric interaction |  |  |

**Complete:** Potential energy describes \_\_\_\_\_; its expression comes from \_\_\_\_\_.

# The Potential-Energy Hinge

Teacher guide

**Purpose.** Students separate the general energy relation  $\Delta E_p = -W_{\text{cons}}$  from the interaction-specific expression and the chosen reference zero. The desired architecture is

interaction law + geometry + reference choice  $\longrightarrow E_p(\text{configuration})$ .

## 1. Near-Earth gravitational energy

$$\Delta E_{p,g} = mg(h_2 - h_1) = 0.40(9.81)(3.5) \approx 13.7 \text{ J}.$$

Work done by gravity:

$$W_g = -\Delta E_{p,g} \approx -13.7 \text{ J}.$$

Moving the zero changes each assigned value of  $E_p$  by the same constant. Energy differences, work and predictions remain unchanged.  $mgh$  assumes an approximately uniform field over a small height range near Earth’s surface. The local value of  $g$  is supplied by the gravitational field.

## 2. Elastic potential energy

$$\Delta E_{p,s} = \frac{1}{2}k(x_2^2 - x_1^2) = 90(0.11^2 - 0.04^2) \approx 0.945 \text{ J}.$$

Work done by the spring:

$$W_s = -\Delta E_{p,s} \approx -0.945 \text{ J}.$$

The quadratic energy follows from the force increasing linearly with extension. Students need only state this qualitatively at IB level. The expression applies while Hooke’s law is valid and the spring remains within its elastic regime.

Roles:

- Hooke’s law supplies the interaction model.
- Energy conservation connects transfers and stored energy changes.
- $x = 0$  fixes the reference configuration.

## 3. Radial fields and signs

With zero at infinity, two separated masses have zero gravitational potential energy. Bringing them together releases energy; the bound configuration therefore has negative potential energy.

For electric interaction:

- like charges:  $Qq > 0 \Rightarrow E_{p,e} > 0$ ;
- unlike charges:  $Qq < 0 \Rightarrow E_{p,e} < 0$ .

For a satellite moving from  $r$  to  $2r$ ,

$$-\frac{GMm}{2r} > -\frac{GMm}{r},$$

so the gravitational potential energy increases: it becomes less negative.

For two positive charges moving apart,  $E_{p,e} = kQq/r$  decreases, so  $\Delta E_{p,e} < 0$ . The electric force does positive work.

The common  $1/r$  dependence comes from radial inverse-square interactions. The sign comes from the attractive or repulsive character of the interaction and the product of the source properties.

## Synthesis answers

| Case                 | Objects                    | Variable       | Reference            | Closure                                   |
|----------------------|----------------------------|----------------|----------------------|-------------------------------------------|
| Near-Earth gravity   | Earth and object           | height $h$     | chosen level $h = 0$ | approximately uniform gravitational field |
| Spring               | spring and attached object | extension $x$  | unstretched spring   | Hooke’s law                               |
| Universal gravity    | two masses                 | separation $r$ | zero at infinity     | Newtonian gravitation                     |
| Electric interaction | two charges                | separation $r$ | zero at infinity     | Coulomb interaction and charge signs      |

Suggested completion:

Potential energy describes the energy of a configuration; its expression comes from the interaction law and the chosen reference.

### Misconceptions to expose

- Potential energy is assigned to the system, not to an isolated object.
- Negative potential energy is meaningful relative to a chosen zero.
- The zero can move; measurable energy differences do not.
- $mgh$  and  $-GMm/r$  describe the same interaction at different levels of approximation.